

Assignment - 01
Artificial Intelligence.

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• Sec-A

1. Give the difference between knowledge and Intelligence.

Ans Knowledge refers to the information, facts, and skills acquired through experience or education. It's what you know.

Intelligence is the ability to learn, understand, and apply knowledge, as well as to reason and solve problems. It's how you use what you know.

2. Explain briefly the difference between procedural and declarative knowledge.

Ans Procedural knowledge is knowing how to do something, like riding a bike or solving a math problem. It involves processes and steps.

Declarative knowledge is knowing facts and information, like knowing that Paris is the capital of France. It involves "what" rather than "how".

3. Explain the types of environment in AI.

Ans In AI, environments are categorized as:

- (i) Fully vs. Partially Observable: Full or partial access to info.
- (ii) Deterministic vs. Stochastic: Predictable or uncertain outcomes.
- (iii) Static vs. Dynamic: Environment stays the same or changes over time.
- (iv) Discrete vs. Continuous: Limited or infinite state/action space.
- (v) Single-agent vs. Multi-agent: One AI agent or multiple interacting agents.

4. What are facets in Frame representation? &

Ans In frame representation, facets are attributes or properties of a slot that provide additional information. They include details such as default values, constraints, or conditions that specify how the slot's value should be used or modified.

5. Discuss the steps needed to solve a problem.

Ans. The steps to solve a problem are:

- (i) Identify the Problem
- (ii) Analyze the Problem
- (iii) Generate possible solⁿ
- (iv) Evaluate Options
- (v) Choose a solution
- (vi) Implement the solution
- (vii) Review and assess.

• Section - A (3X3=21)

(i) Describe the contribution of philosophy and mathematics to Artificial Intelligence.

Ans. Philosophy has contributed to Artificial Intelligence (AI) by addressing fundamental questions about mind, reasoning and consciousness. It explores the nature of knowledge, logic, and ethics, which guide AI's development in terms of reasoning, decision-making and ethical AI system.

Key areas influenced by philosophy -

- Epistemology (study of knowledge) contributes to understanding how AI can acquire, process, and apply knowledge.
- Logic provides the foundation for formal reasoning system in AI, such as inference engines and expert system.
- Ethics is crucial in shaping AI governance, ensuring fairness, accountability and transparency in AI decision-making.

mathematics provides the foundational tools for AI, including algorithms, probability theory, statistics, and logic. It helps in creating models for learning, optimizing, and making decisions. Key branches like calculus, linear algebra, and discrete mathematics enable AI techniques such as machine learning, neural networks, and data analysis.

2. Explain the following terms:

(i) Path cost:- The path cost is the total cost associated with a sequence of actions taken to reach a goal from an initial state in a problem-solving process. It includes the sum of individual step costs, such as distance, time, or resources used.

(ii) Goal state: The goal state is the desired final condition that an AI system aims to achieve when solving a problem. It represents the end of the search or problem-solving process, where the objective is successfully met. In search algo, the goal state defines success.

(iii) Optimal solution to a problem:-

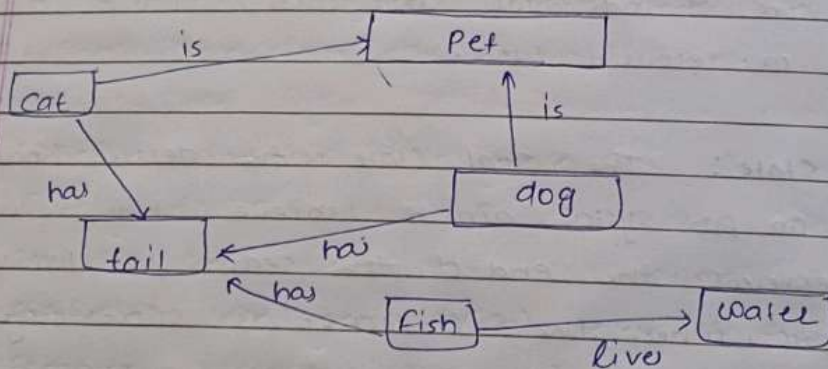
An optimal solution is the best possible solution to a problem, usually the one that minimizes the path cost or maximizes the reward. It ensures the most efficient use of resources while achieving the goal. In AI, an optimal solution balances factors like time, cost, and accuracy.

Q3. What is Semantic Network? Show the semantic network representation with a suitable example.

Ans A semantic network is a knowledge representation technique that represents relationships between concepts in the form of a graph. In this structure, nodes represent entities or concepts, and edges represent relationships between them. It's used to model how knowledge is interconnected, helping machines understand relationships and associations in a logical way.

Example:-

Let's create a simple semantic network to represent the relationships between different animals:-



in above example:-

- Animal is the most general concept.
- mammal and bird are subclasses of Animal.
- Dog and Cat are instance of mammal.
- Parrot is an instance of Bird.
- Tail is a part of a Dog.

- Application of semantic NW:- Artificial intelligence, Natural Language Processing, Information Retrieval, knowledge management.

• Sec-C

3. How to define a Problem as state space search? Discuss it with the help of an example.

Ans: State space search is a framework used in artificial intelligence and computer science to solve problems by exploring possible states and actions in a structured way.

To define a Problem as a state space search, follow these steps:

- (i) State Space - Identify all possible configurations or states in the problem domain.
- (ii) Initial state - Determine the starting configuration or situation.
- (iii) Actions :- Define the set of possible actions or moves that transition from one state to another.
- (iv) Transition Model - Describe how each action changes the state.
- (v) Goal State(s) - Specify the target configuration(s) that represent a solution.
- (vi) Path Cost - (if applicable) Define the cost associated with transitioning between states.

Example 1: The 8-puzzle Problem -

- (i) State space - All possible configurations of the 3x3 grid with tiles 1-8 and a blank space (0).
- (ii) Initial State - A scrambled grid, e.g:

1	2	3
4	5	6
7	8	0

- (iii) Action - Moves that slide the blank space (0) up, down, left, or right.

(iv) Transition Model - Moving the blank space results in a new grid configuration. E.g., moving 0 left.

1	2	3
4	5	6
7	0	8

(v) Goal State :- The grid arranged in numerical order with the blank space in the bottom-right corner.

1	2	3
4	5	6
7	8	0

(vi) Path Cost :- Number of moves needed to reach the goal state from the initial state.

Q2. What is "Artificial Intelligence" and "AI Techniques"?

Briefly explain how AI Techniques can be represented and list out some of the task domain of AI.

Ans Artificial Intelligence (AI) refers to the field of computer science dedicated to creating system or machines capable of performing task that typically require human intelligence. These task include problem-solving, reasoning learning, and understanding natural language.

Artificial Intelligence Techniques are specific methods or approaches used within AI to achieve intelligent behavior. These techniques are employed to create train, and utilize AI system to perform various tasks effectively.

• Representation of AI Techniques:-

AI techniques can be represented in various ways, depending on the nature of the task and the problem being solved. Common representations includes-

- (i) Algorithms - Step-by-step Procedures or formulae for solving specific problem.
- (ii) Models - Abstract representations of real-world process or system (ex - neural NW, decision making, etc.).
- (iii) Rules - Set of conditional statement that decides action based on input.
- (iv) Graph - Structure used to represent relationship and pathways.

• Task Domain of AI:-

AI can be applicable to a wide range of task domain, including -

- (i) Natural Language Processing (NLP) - Understanding and generating human language (ex - chatbot lang. translation).
- (ii) Computer vision - Interpreting visual information from the world (ex - image recognition, object detection, etc.).
- (iii) Robotics - Designing and operating robots to perform tasks autonomously. ex (autonomous vehicles).
- (iv) Machine Learning - Training systems to improve their performance on task through experience.
- (v) Decision making - Making decisions based on data and predefined criteria (eg - credit scoring.)

all over all, AI techniques are methods used to create intelligent systems and can be represented through algo, model, rules, graphs, and matrices.

Q3. Discuss various approaches and issues in knowledge representation. Also discuss various problems in representing knowledge.

Ans. Knowledge representation is a crucial aspect of AI and involves encoding information about the world in a format that a computer system can utilize to solve complex problems. Effective knowledge representation allows AI systems to reason, make decisions, and understand context.

• Approaches to Knowledge Representation -

(i) Logic-Based Approaches -

- Propositional Logic - Simple true/false statements. Limited in expressing relationships.
- First-Order Logic - More expressive, allows for objects and relationships.

(ii) Semantic Networks -

- Conceptual Graphs:- Nodes represent concepts, edges represent relationships.
- Frames - Data structures with attributes and values for representing knowledge.

(iii) Production Rules -

- Rule-Based Systems - "If-then" for decision making.

(iv) Ontologies -

- Ontology-Based Representation - Formal structure including classes and relationships. Useful for domain-specific knowledge.

(V) Frames and Scripts -

- Frames - Hierarchical data structure representing stereotypical situations or objects. Frames capture both the static and dynamic aspects of knowledge.

• Issues in Knowledge Representation -

(i) Expressiveness -

- Complexity - Difficulty in representing complex or nuanced knowledge.
- Scalability - challenge in managing large volumes of knowledge.

(ii) Consistency -

- Conflict - Ensuring no contradictory information.
- Maintenance - keeping the knowledge base up-to-date.

(iii) Relevance -

- Overhead - Avoiding irrelevant or excessive details.
- Context Sensitivity - Adapting knowledge to specific context.

(iv) Uncertainty -

- Handling Uncertainty - Representing and reasoning with incomplete or uncertain information.

~~Problem~~

- Problem in Representing Knowledge -

(i) incompleteness: knowledge base may lack n

need